Disc Harrow



Category: **Agriculture & Gardening** Type: **Full SOP** Priority: **High** 7 labs

Covers these lab items: Disc Harrow

What it is used for

A disc harrow breaks soil clods, cuts crop residue, and helps prepare fields after ploughing. Correct tractor attachment, depth adjustment, and safe transport reduce injury and equipment damage.

Main parts

- Disc blades cut and turn the soil.
- Disc gangs hold multiple blades at the required angle.
- Frame supports the disc gangs and connecting components.
- Hitch or linkage connects the harrow to the tractor.
- Scrapers help remove soil and residue from the discs.
- Depth or angle adjustment controls working depth and soil action.
- Transport wheels or supports help move the implement where fitted.

Before you start

1. Read the operator's manual and use the implement only after suitable training.
2. Inspect discs, bearings, frame, hitch, fasteners, scrapers, and transport components for damage.
3. Confirm the tractor is compatible with the harrow's size, weight, and hitch requirements.
4. Attach the harrow on firm, level ground with the tractor secured against movement.
5. Ensure hitch pins and locking devices are correctly installed before moving.
6. Clear people, animals, stones, wires, and other hazards from the field.

How to use

1. Set the tractor parking brake and lower or support the implement safely before adjustments.
2. Connect the harrow using the approved hitch and secure all locking devices.
3. Adjust disc angle and working depth according to the soil and operating instructions.
4. Raise the implement to the transport position before travelling between work areas.
5. Move slowly to the field and lower the harrow only after aligning with the work direction.
6. Engage the tractor drive and operate at the recommended speed for the soil conditions.
7. Maintain straight, controlled passes and monitor soil coverage, vibration, and implement movement.
8. Raise the harrow before turning sharply, reversing, or crossing unsuitable ground.
9. Stop the tractor, secure it, and lower or support the implement before inspection or clearing residue.
10. After work, remove accumulated soil, inspect for damage, and park the harrow securely.

⚠ Safety

- Never stand beneath a raised harrow or between the tractor and implement during hitching.
- Keep hands away from sharp disc edges and moving linkage points.
- Stop the tractor engine and secure the machine before adjusting, cleaning, or repairing the harrow.
- Never carry passengers on the implement or allow bystanders near operating discs.
- Use suitable gloves and footwear when handling stationary discs, while keeping gloves away from moving machinery.
- Follow tractor and implement limits on slopes, speed, and transport.

Care & storage

- Clean soil and crop residue from discs and scrapers after safely stopping the equipment.
- Inspect disc edges, bearings, hitch pins, fasteners, and frame for wear or damage.
- Lubricate bearings and moving joints according to the manufacturer's maintenance schedule.
- Store the harrow on firm, level ground with the implement secured against movement.

Common mistakes to avoid

- Adjusting the harrow while it is raised without proper support.
- Travelling with the implement lowered or at an unsafe speed.
- Ignoring worn discs, loose hitch pins, or damaged bearings.
- Standing between the tractor and harrow without securing the tractor against movement.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar